The Data-First Study: do patterns found in the earthquake data match the Vedic rules?

In plain words. The companion Rule-First Study takes rules from the old texts and checks them against earthquakes. This study works the other way round. It starts with the earthquakes, asks which positions of the Sun, Moon and planets turn up unusually often when an earthquake strikes, and only afterwards compares what it finds with the texts. It has five stages: a scan of the Modern M6.5+ set (1,270 earthquakes), a repeat on the much larger Modern M5.5+ set (9,368), tests of combinations of conditions that the first stages left out, a repeat on older earthquakes (1900–1989), and a scan of every bin that the source texts themselves define (Parāśara Tantra, Brihat Saṃhitā in Sanskrit and English, Adbhuta Sāgara). The earlier stages scanned generic sky bins (every nakṣatra, sign, tithi and planet separation), which already contain the nakṣatra circles of every treatise; stage 5 adds the bins that need a treatise to define them. Technical terms are kept where they help and explained in the Terms table. Report date: 20 October 2026.

Sources. The Vedic rules that the data are compared with come from four source texts: the Parāśara Tantra, the Brihat Saṃhitā, the Adbhuta Sāgara and Cunningham's Ancient Geography of India (regions), with Madhavan's Muddle of Ayanamsa and the TG-SAI repository as supporting tools. The Rule-First Study (section 3) lists every rule by source.

The answer in brief

No star group, Moon phase, planetary arrangement, mandala or combination, from any of the source texts, was reliably linked to earthquakes. The two leads from the first scan vanished when tested on fresh earthquakes, and the one faint lead that remained did not replicate on older data.

Question	Answer	Where
Does a scan of 433 sky bins on the Modern M6.5+ set find a reproducible pattern?	Only faintly (r=0.105 between the two halves). The top bin, the Sun in Viśākhā, did not replicate on 1964–1989 (1.15×, p=0.24).	Section 2
Do the leads survive on 8,098 new earthquakes (Modern M5.5+ set)?	No. Moon in the sharp group 1,176 vs 1,199 expected (p=0.77); Sun in Viśākhā 291 vs 287 (p=0.41).	Section 3
Does any of the 23 rules of the Rule-First Study pass on the larger set?	No. The closest (Moon within 5° of the Sun or a planet) had p=0.0030 against a bar of 0.0022.	Section 3.4
Do untested combinations and slow-planet factors pass?	No. 20 primary tests (bar 0.0025) and 91 pairs (bar 0.00055) all fail.	Section 5
Do the bins defined by the Brihat Saṃhitā, Adbhuta Sāgara and Parāśara Tantra (54 bins) show a pattern that repeats?	No. Six bins reached p<0.05 in 1990–2007 (all the same Moon-in-Indra-circle deficit under six lists); none held in 2008–2026 (z between −0.2 and 0.0).	Section 7
Does the closest lead replicate on 1900–1989 earthquakes?	No (0.98×, p=0.73); the scan and the TG-SAI model also show nothing.	Section 6

1. Idea, approach and data sets

Testing the texts' rules one at a time depends on how each rule is read (which nakṣatra, which window, which threshold). Starting from the data avoids that choice: every available sky variable is cut into bins in the same way, so no rule has to be interpreted first. The cost is many more comparisons. That is handled by splitting the data into two halves (discovery, before 2008; holdout, from 2008), by a single summary statistic, and by testing every lead on fresh data.

Data sets (names used in both reports)

Name	Years	How selected	Events	Used for
Deadly 15	1999–2026	The 15 deadliest earthquakes in the USGS catalogue (file usgs_catalogue_15_events.csv)	15	First tests of every rule
Modern M6.5+ set	1990–2026	All M6.5+ earthquakes, aftershocks removed (1,709 raw)	1,270 (632 before 2008 = discovery; 638 from 2008 = holdout)	Pre-registered 23 rules; first data-first scan
Modern M5.5+ set	1990–2026	All M5.5+ earthquakes, aftershocks removed; includes the Modern M6.5+ set	9,368 (4,661 before 2008; 4,707 from 2008); 8,098 are not in the Modern M6.5+ set	Larger-sample replication; gap tests
Older set 1	1964–1989	M5.5+, aftershocks removed (11,329 raw)	6,111	Replication on never-used data

Name	Years	How selected	Events	Used for
Older set 2	1900–1963	M7.0+, aftershocks removed (759 raw)	670	Replication on never-used data
Repository file	1850–1950	Table of historical M5+ quakes shipped with the TG-SAI repository	5,003	Scoring the TG-SAI model
Aftershock set	1990–2026	481 main shocks of M7.0+ with USGS aftershocks (M5.0+, 100–300 km, 0–90 days)	481 sequences (11,608 records)	Verse 18.7 robustness test

The four stages

Stage	Question	Data	Section
1	Does a scan of every sky bin show a reproducible pattern?	Modern M6.5+ set; follow-up on the M6.5+ part of Older set 1	2
2	Do the leads, and the 23 rules, hold on a sample 7.4 times larger?	Modern M5.5+ set (8,098 events never used before)	3
3	Do combinations of conditions and slow-planet factors show anything?	Modern M5.5+ set	4, 5
4	Does the closest lead replicate on older earthquakes?	Older set 1, Older set 2	6

2. Stage 1: the scan of 433 sky bins on the Modern M6.5+ set

Step	What was done	Why
1. Data	The Modern M6.5+ set: USGS worldwide earthquakes M6.5+, 1 Jan 1990 to 19 Oct 2026, 1,709 events reduced to 1,270 independent main shocks by removing aftershocks (Gardner–Knopoff windows).	Aftershocks cluster in time and would inflate any pattern.
2. Sky positions	The Sun, Moon and five planets for each event from the ephemeris tool (Lahiri sidereal), plus retrograde flags.	One consistent source for every variable.
3. Background	A 6-hourly grid of sky positions over 1990–2026, built the same way: the picture of what a random moment looks like.	Expected counts must come from real sky states.
4. Binning	Every variable cut into bins: 433 in all (table below).	Every variable is treated alike; no rule is chosen first.
5. Compare each bin	For each bin, events seen vs expected from the background, as a z-score, separately for the discovery half (before 2008, 632 events) and the holdout half (2008 onward, 638 events). The background is matched to each half's years.	Slow planets occupy different states in different decades, so a pooled background would mislead.
6. Reproducibility test	Correlate the 433 discovery z-scores with the 433 holdout z-scores (r). Compare with 300 chance versions in which events were drawn from the background.	If real structure exists, bins high in one half are high in the other. One number, so no search over bins is needed.
7. Follow up the top bin	Freeze one rule for the strongest bin, write it down, and test it once on an independent catalogue (1964–1989).	The only fair test of something found by looking.
8. Audit	Rerun the Rule-First Study's slow-planet rules with a background matched to each half.	A flaw in the baseline became visible while building step 5.

What was binned

Family	Bins	Count
Nakṣatra of the Sun, Moon, Mars, Mercury, Jupiter, Venus, Saturn	27 each	189
Sidereal sign of the same seven	12 each	84
Moon's tithi (from the Moon–Sun angle)	30	30
Moon's angle from each of five planets	12 bins of 30° each	60
Sun's angle from each of five planets	12 bins of 30° each	60
Retrograde yes/no for five planets	2 each	10
Total		433

Results

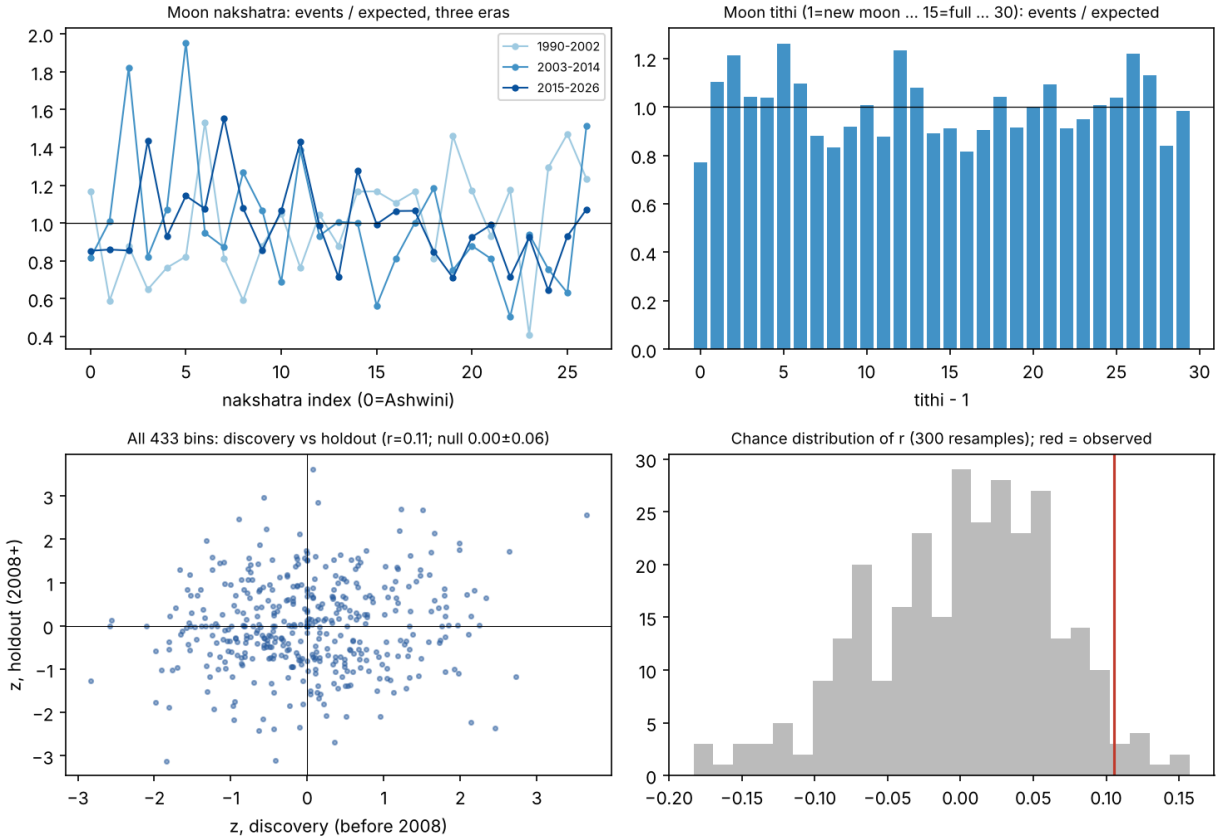


Figure 1. Top left: events divided by expected for the Moon's nakṣatra in three eras; the lines disagree, which is what noise looks like. Top right: Moon's tithi (1 = new moon, 15 = full moon); no consistent excess. Bottom left: all 433 bins, discovery z against holdout z. Bottom right: the observed r (red line) against 300 chance versions.

Reproducibility. r=0.105; chance mean 0.00, spread 0.064 (p=0.03 for this single summary number). With the most extreme bin removed, r=0.086 (about 1.3 spreads, p≈0.10); with every Sun-related bin removed, r=0.078. The evidence for structure is at best faint.

Ten bins with the largest discovery z. Several of the biggest discovery values shrank or reversed in the holdout, which is the usual behaviour of chance extremes.

Bin	z, 1990–2007	z, 2008–2026
Sun in Vishakha	+3.65	+2.57
Sun in U.Phalguni	-2.83	-1.27
Saturn in P.Bhadrapada	+2.73	-1.17
Moon 30-60 deg ahead of Mercury	+2.64	+1.71
Saturn in sidereal Aries	-2.58	+0.00
Jupiter in P.Bhadrapada	-2.56	+0.14
Sun in Mrigashira	+2.46	-2.37
Venus in Bharani	+2.33	+0.65
Sun in Hasta	+2.25	+0.02
Mercury in sidereal Aries	+2.18	+0.81

The one standout: the Sun in Viśākhā

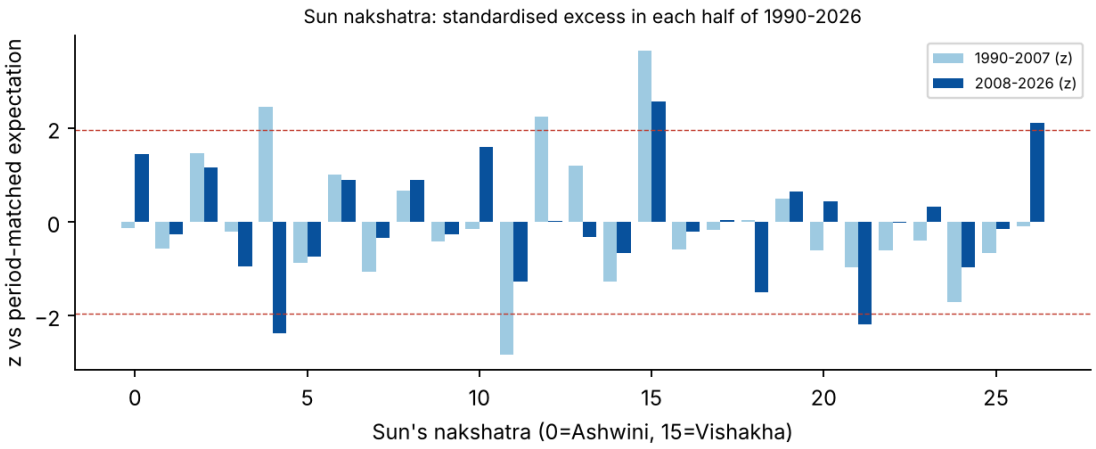


Figure 2. Standardised excess for the Sun's nakṣatra in each half; the Viśākhā bar (index 15) is high in both. It covers about 16–29 November.

The Sun in Viśākhā (sidereal 200°–213.3°) accompanied 74 events against 45.0 expected (1.64×). It appeared in all three eras (26 vs 16.7, 27 vs 15.6, 21 vs 12.8), more strongly in M7.0+ events (33 vs 17.0) and shallow quakes (62 vs 32.9), and with no excess in deep ones (12 vs 12.1). Neither text gives the Sun in Viśākhā a role in earthquakes; the Parāśara Tantra pairs Viśākhā with Jupiter and Saturn in a conjunction rule, and the Brihat Saṃhitā places it in the Agni (fire) mandala.

Confirmation on independent data

Before fetching new data we wrote down one test (file preregistration_bottom_up_followup.md): the same Sun window, one-sided, $p<0.05$ required, on USGS M6.5+ earthquakes from 1964 to 1989, declustered the same way (the M6.5+ part of Older set 1: 990 events, 768 main shocks), with the Sun's position from a standard solar formula that matched the ephemeris tool within 0.013° on the 1990+ events. **Result:** 32 observed vs 27.9 expected (1.15×, $p=0.24$); in M7.0+ events 6 vs 9.4. The excess did not replicate and the finding is treated as a likely chance cluster. By itself a bin reaching $|z|=3.65$ somewhere among 433 bins happens in about one scan in ten ($433 \times 2.6\times10^{-4}$), and the bins overlap (Sun nakṣatra, sign and separations share the same dates), so its repeat in both halves and all eras was less unusual than it looked.

Audit of the baseline

Building step 5 showed that the Rule-First Study's 23-rule test had used a background pooled over all years. For slow planets this can bias the p-value, because their positions differ between decades. The affected rules were rerun with a background matched to each half. No conclusion changed.

Rule	Holdout observed vs expected	Holdout p (matched)
R7 Mars retrograde	63 vs 57.9	0.259
R7 Mercury retrograde	138 vs 122.4	0.066
R7 Jupiter retrograde	169 vs 189.3	0.965
R7 Venus retrograde	41 vs 45.6	0.782
R7 Saturn retrograde	252 vs 240.7	0.188
R8 Jupiter=Saturn nakshatra	20 vs 23.2	0.779
R9 planet pair within 3 deg	134 vs 133.8	0.506

None approaches the corrected threshold of 0.0022 (0.05 divided by 23 rules). Fast-moving variables (Moon, tithi, Sun) are unaffected by the change.

Stage 1 in one paragraph. The data-first scan agrees with the rule-first view: no nakṣatra, tithi, planetary state or mandala is reliably over-represented, so none of the mandala schemes (Parāśara Tantra, Brihat Saṃhitā, Adbhuta Sāgara) finds support. The Sun-in-Viśākhā lead is in none of the texts and did not hold up. The agreement between the halves ($r=0.105$ against a chance spread of 0.064) was faint, and stage 2 tests whether it was real.

3. Stage 2: the Modern M5.5+ set (9,368 earthquakes)

The Modern M6.5+ set is large enough to catch big effects but could miss small ones, and the leads found in it (the Sun in Viśākhā, and the Moon in the 'sharp' group) might have been luck. So every smaller earthquake of magnitude 5.5 and above was added and the work repeated on a sample 7.4 times larger. Where this stage

and stage 1 differ, rely on this stage.

Step	What we did	In plain terms
1. Freeze the plan	Wrote down the two main tests and the pass mark before looking at any new data (file preregistration_m55_replication.md).	Deciding the rules of the game before playing, so they cannot be adjusted to suit the result.
2. Get more earthquakes	Downloaded all earthquakes of magnitude 5.5 to 6.5 for 1990–2026 from the US Geological Survey, added the Modern M6.5+ set, and removed aftershocks. Result: 9,368 independent earthquakes, of which 8,098 are new.	Smaller quakes are far more common. Only the 8,098 new ones serve as a fair test, because the older ones helped suggest the leads.
3. Sky positions	The ephemeris tool was offline, so Sun, Moon and planet positions were computed with our own formulas and checked against the tool on the Modern M6.5+ set: Moon and Sun agree to 0.01°, planets within 0.17° at worst, and all retrograde flags are identical.	A star group is 13° wide, so an error of a fraction of a degree changes almost nothing.
4. Primary tests	Is the Moon in the sharp group more often than chance on the 8,098 new events? Is the Sun in Viśākhā?	One question each, one chance to answer it.
5. Rerun the 23 rules	All 23 rules from the Rule-First Study on all 9,368 events, with expected counts matched to the years of each half.	A full repeat of the first experiment at a larger scale.
6. Redo the scan	The 433-bin scan: do the bins that stand out in the first half stand out in the second?	A repeat of stage 1.
7. Extra checks	Results by earthquake size, and a 'time-shift' test that keeps earthquake clusters intact but breaks any link with the sky.	To see whether results depend on size and that leftover clustering does not fake a result.

3.1 The two main tests

Test	Events	Seen	Expected	Seen ÷ expected	p-value	Verdict
Moon in sharp group	8,098	1176	1199.2	0.98	0.77	Not confirmed
Sun in Vishakha	8,098	291	286.8	1.01	0.41	Not confirmed

p-value (plain meaning): the chance of seeing a result this strong if sky and earthquakes were unrelated. A large value such as 0.77 means the data look just like chance. We had said in advance that a p-value below 0.025 in the expected direction was needed. **Size of effect that is now excluded:** for the sharp group the 95% range is 0.93× to 1.03×, so an excess of 17% (the earlier figure) is ruled out, and with 8,098 events an excess of about 10% or more would reliably have been noticed.

3.2 How the leads behave by earthquake size

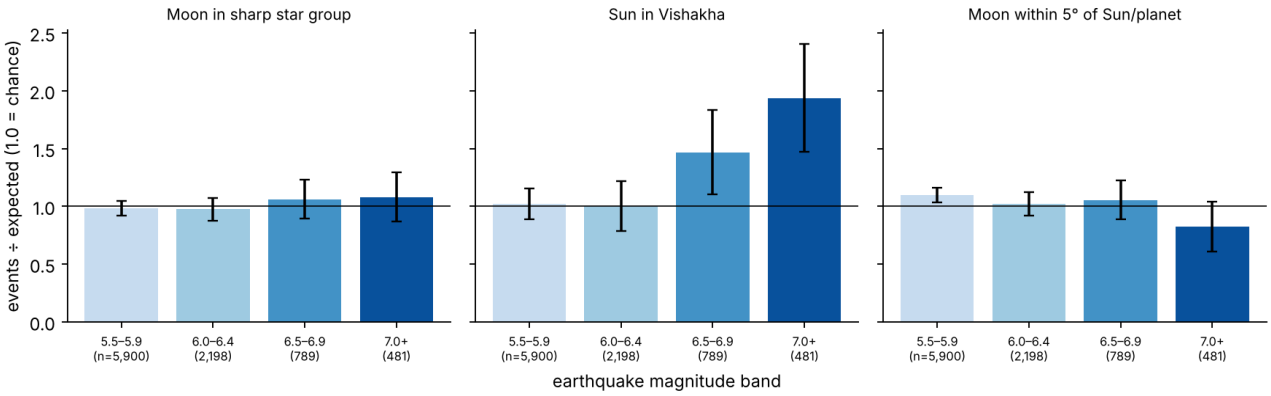


Figure 3. Each bar is the number of earthquakes divided by the number expected from chance (1.0 = no link); the black line shows the uncertainty. **Left:** the sharp group sits at chance for small quakes and slightly above 1 for large ones, but within the uncertainty. **Middle:** the Sun in Viśākhā is at chance for 7,000+ smaller quakes and high only in the 1,270 largest, which are the very ones where it was first noticed. **Right:** the Moon near the Sun or a planet is slightly high for the smallest quakes only.

Reading the Viśākhā result. A real physical link would not be expected to switch on only in the group that suggested it. The pattern is typical of a coincidence found by searching: it looked strong in the sample used to find it and vanished in a new sample 6.4 times larger. A caution remains: among the largest quakes (M7+) it stays high (33 vs 17.0, p≈0.006 even after accounting for clustering), but the older 1964–1989 catalogue did not show it (6 vs 9.4), so it is not confirmed.

3.3 The 23 rules rerun

All 23 rules were repeated on the 4,707 events from 2008 onward (with 4,661 before 2008 for comparison). The pass mark is 0.0022: with 23 tests, about one would dip below 0.05 by luck, so each must reach the stricter 0.05 ÷ 23 (the Bonferroni correction). **No rule passed.** The six smallest p-values were:

Rule	Seen vs expected	Ratio	p-value
R10 Moon within 5 deg of planet/Sun	756 vs 688.2	1.099	0.0030
R12 mandala vs mag [sid]	H=7.44		0.0592
R5 eclipse within +/-15 d	1305 vs 1265.5	1.031	0.1001
R6 eclipse within +/-7 d	835 vs 804.7	1.038	0.1242
R9 planet pair within 3 deg	1019 vs 986.8	1.033	0.1286
R4 within +/-1 tithi of new/full moon	955 vs 924.7	1.033	0.1376

The mandala schemes of the Parāśara Tantra and the Brihat Saṃhitā again show an even spread (new events: p=0.28 and p=0.12; the Adbhuta Sāgara lists were tested later and are also within chance, section 7), and the sharp-group, tithi, eclipse and retrograde rules are all within chance. The mandala–magnitude link sits at p=0.059.

3.4 The closest call: the Moon near the Sun or a planet

Rule 10 asked whether the Moon is within 5° of the Sun or a planet more often than expected. On 2008-onward events it gave 756 against 688 (1.10×, p=0.0030), just missing the 0.0022 pass mark; before 2008 it was weaker (1.03×, p=0.25). Splitting by size (an after-the-fact look) shows the excess only in the smallest quakes (M5.5–5.9: 950 vs 866, 1.10×) and absent in the largest (0.82× at M7+). The two parts of the rule behave similarly (Moon near the Sun about 1.13×; Moon near a planet about 1.12×; strongest for Jupiter 1.17× and Venus 1.14×). The time-shift test, which keeps clusters of quakes intact, gives p=0.007 for all new events and 0.002 for the M5.5–5.9 band, so clustering does not explain it. **How to read this:** with 23 rules, the smallest p-value drops to 0.003 about 7 times in 100 by luck alone, and the size split was chosen after the fact. It is suggestive, runs the wrong way for large quakes, and is not a finding. The fresh test was then run on 1964–1989 and 1900–1963 data, and the lead did not replicate (section 6).

3.5 The scan, redone

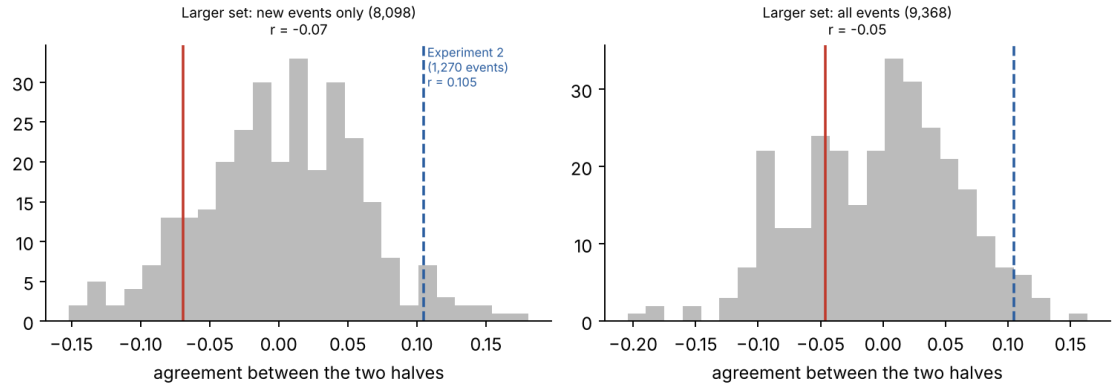


Figure 4. The grey bars show how much agreement between the two halves luck alone produces; the red line is what we saw (the blue dashed line shows the stage 1 result on the Modern M6.5+ set (1,270 events)). In stage 1 the agreement was faintly above chance (r=0.105); with the larger set it is slightly below zero (r=−0.07 for the new events, −0.05 for all events). The stage 1 'trace of structure' was therefore noise. The largest single bin is now the Sun 30–60° ahead of Mars (z=3.95 in the first half, 0.76 in the second), which does not repeat.

What this means. Seven times more data gives the same answer as before: no star group, Moon phase, planetary arrangement or mandala from the Parāśara Tantra, the Brihat Saṃhitā or the Adbhuta Sāgara is reliably linked to earthquakes of magnitude 5.5 and above. The two leads were chance clusters. 'Nothing found' means nothing at this level of detail: effects under about 10% for groups of this size, or effects tied to a particular region or time of year that were not searched, cannot be excluded.

4. Scope: what was covered and what was not

The table shows each kind of sky factor the texts mention, and whether it was tested on the Deadly 15 and on the larger sets. **Covered** means a rule was tested; **Gap** or **Partly** means some or all of it has no test. A gap is not a verdict that the factor is unimportant. The factors marked as gaps in stage 2 are the subject of section 5.

Sky factor (source chapter)	Deadly 15	Larger sets (Modern M6.5+ and M5.5+)	Status
Moon's star group → earthquake type (PT 18.3–18.6; BS 32.8–22)	Tested	Tested (R1; Brihat Saṃhitā version too)	Covered
Moon in a 'sharp' star group (PT ch.14)	Tested	Tested (R2, and the Data-First Study lead)	Covered
Moon's phase: tithi, karaṇa, new/full moon (PT ch.16)	Tested	Tested (R3, R4; 30 tithi bins)	Covered
Time of day (PT 18 quarters; BS 32.3–7 halves)	Tested (3 of 15 matched)	Brihat Saṃhitā version only (152 vs 159)	Partly
Sun's season: six ṛtus (PT ch.3)	Tested	Tested (R11)	Covered
Sun's star group and sign	Tested (season only)	Tested in the data-first scan (27 + 12 bins)	Covered
Eclipses (PT ch.5)	Tested (±7, 15, 30 days; 24-month rule; regions)	±7 and ±15 days only (R5, R6)	Partly: regions and 24-month rule not rerun
Comets (PT 11.19)	Tested (7 great comets)	Not rerun	Low power
Planet retrograde: Mars, Mercury, Jupiter, Venus, Saturn (PT ch.6–10)	Tested	Tested (R7)	Covered
Each planet's own star group and sign	Mars and Venus partly	Tested in the data-first scan (all 5 planets)	Covered
Moon close to the Sun or a planet (PT ch.14)	Tested (within 5°)	Tested (R10; Moon–planet angles in 30° steps)	Covered; R10 near miss did not replicate pre-1990
Sun at an angle to each planet	Not tested	Tested in the scan (30° steps)	Partly
Any two planets within 3° (PT ch.15 'war')	Tested (1° and 3°)	Tested (R9)	Covered
Jupiter and Saturn in one star group	Tested	Tested (R8)	Covered
Three or more planets in one star group (PT ch.15 formations)	Tested (0 of 15)	Tested (G1, both frames)	Covered (not confirmed)
Venus's six circles (PT ch.9)	Tested	Tested (G2, both frames)	Covered
Mars stationary; Saturn in Kṛttikā with Jupiter in Viśākhā	Tested	Mars stationary tested (G3); the Saturn–Jupiter arrangement did not occur in 1990–2026	Covered / cannot be tested
Planet-to-planet angles other than 'close' (90°, 120°, 180°)	Not tested	Tested (G4: opposition, square, trine)	Covered; specific pairs not tested
Rāhu and Ketu positions	Not tested (eclipses used instead)	Tested (G5: Rāhu's star group; Moon near the node axis)	Covered
Moon sharing a star group with planets (G6); a planet next to a particular named star	Not tested	Moon with planets tested (G6); named stars not tested (star positions not available)	Partly
Combustion (planet close to the Sun); planet rising or setting	Counted in 15 events	Combustion tested (G7); rising and setting beyond that not tested	Partly
Combined conditions (for example sharp-group Moon while Saturn is retrograde); an 'affliction score'	Not tested	Tested (G9 score; 91 two-condition pairs)	Covered
TG-SAI physics features (lunar perigee, zenith, Moon–Sun alignment)	Tested	Retrained on the Modern M6.5+ set (AUC about 0.50)	Covered
Aftershock recurrence (PT 18.7)	Tested (15 events)	Tested (481 M7+ sequences)	Covered
Omens and effects in the texts (smoke, meteors, haloes, weather, disease, regions)	Not testable	Not testable	No records exist

What this means for the conclusions. 'No pattern found' applies to the factors listed as Covered. The texts often describe effects of several conditions together; those combinations, Rāhu and Ketu, wider planet angles, combustion and the Ch.15 formations are the subject of section 5. Each added test makes a lucky

result more likely, so they were run with rules written in advance and a stricter pass mark.

5. Gap tests on the Modern M5.5+ set

After the scope table showed gaps, we wrote down rules for the gap factors that can be tested (file preregistration_gap_tests.md) and ran them on the 9,368 earthquakes with the same before-2008 / 2008-onward split. **Family:** 20 primary tests, so the pass mark is 0.05 ÷ 20 = 0.0025, and the pattern must also point the same way before 2008. A second family of 91 two-condition combinations (all pairs of 14 base conditions, 'A and B at the same time') has a stricter mark of 0.05 ÷ 91 = 0.00055. In plain terms: more tests, more chance of a lucky result, so the bar is higher.

Gap test	2008+: seen vs expected	Ratio	p-value	Before 2008: ratio, p	Verdict
G1 formation 3+ planets in one nakshatra [sid]	191 vs 162.0	1.18	0.0128	0.93, p=0.847	No
G2 Venus circle [sid]	726/668/670/724/855/...	1.04	0.9688	1.03, p=0.838	No
G1 formation 3+ planets in one nakshatra [trop]	176 vs 165.6	1.06	0.2150	1.05, p=0.257	No
G2 Venus circle [trop]	596/760/706/756/801/...	1.06	0.3155	1.10, p=0.081	No
G3 Mars stationary	118 vs 90.1	1.31	0.0026	0.91, p=0.821	No
G3 Saturn in Krittika & Jupiter in Vishakha	0 vs 0.0		1.0000	never occurred	No
G4 any planet pair within 3 deg of opposition	463 vs 457.9	1.01	0.4085	0.99, p=0.644	No
G4 any planet pair within 3 deg of square	1238 vs 1252.0	0.99	0.6828	1.03, p=0.142	No
G4 any planet pair within 3 deg of trine	1108 vs 1123.7	0.99	0.7092	1.02, p=0.183	No
G5 Rahu nakshatra	205/170/189/166/178/...	1.19	0.0188	1.28, p=0.001	No
G5 Moon within 12 deg of node axis	631 vs 630.3	1.00	0.4953	1.02, p=0.296	No
G6 Moon in same nakshatra as any of 5 planets	821 vs 769.2	1.07	0.0224	1.03, p=0.164	No
G6 Moon in same nakshatra as Mars or Saturn	359 vs 337.2	1.06	0.1145	1.00, p=0.497	No
G7 Mars combust	734 vs 760.5	0.96	0.8579	0.93, p=0.978	No
G7 Mercury combust	2066 vs 2034.0	1.02	0.1774	1.01, p=0.292	No
G7 Jupiter combust	351 vs 343.5	1.02	0.3446	0.95, p=0.832	No
G7 Venus combust	741 vs 741.5	1.00	0.5147	1.03, p=0.172	No
G7 Saturn combust	425 vs 419.1	1.01	0.3885	0.91, p=0.982	No
G7 two or more planets combust	1014 vs 1015.2	1.00	0.5231	0.96, p=0.910	No
G9 affliction score (mean)	1.3327 vs 1.3268	1.00	0.3496	0.99, p=0.702	No

Result: none of the 20 primary tests passes. The closest are Mars standing still (118 vs 90.1, p=0.0026, just above the mark, and the pattern points the other way before 2008) and the Moon sharing a star group with a planet (p=0.022). The affliction score, which adds up nine harmful conditions per earthquake, averages 1.333 against 1.327 expected: no difference. Saturn in Kṛttikā with Jupiter in Viśākhā never occurred during 1990–2026, so it cannot be tested with this period of data.

Two-condition screen. Of 91 pairs, none passes. The strongest was 'Moon within 5° of the Sun or a planet' together with 'a planet combust' (505 vs 442.7 expected, p=0.0012 against the 0.00055 mark, and only 1.05× before 2008). It rests on the Moon-near-planet condition that was already the near-miss in section 3.4.

A caution on slow planets. The p-values above treat each earthquake as independent. For slow-moving points such as Rāhu (18.6-year cycle) or planet formations, that overstates significance, because the sky changes little from one earthquake to the next. Rāhu's star group first looked striking before 2008 (p=0.0005, chi-square), but when the earthquakes were all shifted by a random number of years (which keeps their clustering and breaks any link with the sky), chance produced larger values than the real data in 78% of runs (p=0.78). Formations (3+ planets in one star group) give p=0.037 (star-based) and 0.018 (seasonal) under the same check, but fail the before-2008 requirement. This shift check is saved in gap_slow_planet_time_shift_check.csv; for the earlier slow-planet rules (retrogrades, Jupiter–Saturn) the conclusions are unchanged because none came near significance.

What is still not tested: a planet next to a particular named star (star positions not available here), planet rising and setting beyond the combustion limits, and the specific planet pairs one by one. The omens and effects in the texts remain untestable because no event records them.

6. Stage 4: older earthquakes, 1900–1989

Why we did this. The one faint lead in the 1990–2026 data was the Moon sitting within 5° of the Sun or a planet in small earthquakes (p=0.0030, just short of the 0.0022 bar). A lead found by searching can only be trusted if it appears again in data that was never used to find it. Earthquakes before 1990 are exactly that. We wrote the rules down first (file preregistration_historical_test.md), then ran them once. **Data.** USGS's catalogue, which includes the reviewed ISC-GEM records, for 1964–1989 (M5.5 and above, 11,329 raw events, 6,111 after removing aftershocks) and for 1900–1963 (M7.0 and above, 759 raw, 670 after declustering). The two sites that hold other historical catalogues (NOAA and ISC) cannot be reached from this workspace. Sky positions come from our own ephemeris, validated earlier, with a matching six-hourly background for each era. **Result.** The lead disappears: 581 events against 594.6 expected (0.98×, p=0.73), the wrong direction. The same holds for all sizes, for the earlier era and when dates are shifted to keep clusters intact. None of the other 21 rules passes, and neither does the Sun in Viśākhā or the Moon in the sharp group. The lead is therefore closed as a chance cluster. **Two honest notes.** (1) Jupiter and Saturn sharing a star group showed fewer events than expected in both eras (168 vs 213 and 24 vs 31). That is a shortfall, not the excess the rules predict, and slow planets repeat their positions for years, so it most likely reflects the timing of a few conjunctions counted many times, not a real effect. (2) Depth is missing from the 1900–1963 records, so the depth tests could not be run for that era. Eclipse rules were not run for either era.

Era (events after declustering)	Test (frozen in advance)	Observed vs expected	p-value	Result
1964–1989, M5.5+ (6,111)	Moon within 5° of Sun or a planet, M5.5–5.9 only (the single primary test)	581 vs 594.6 (0.98×)	0.73 (bar 0.05)	Not replicated
1964–1989, M5.5+ (6,111)	Same rule, all sizes / M6+	877 vs 898.9 (0.98×); 296 vs 304.3 (0.97×)	0.79 / 0.71	Not replicated
1964–1989	Time-shift check of the primary test (keeps clusters intact)	0.98× the shifted-date average	0.69	Not replicated
1964–1989	The 21 other rules (eclipse rules not run), bar 0.0024	Smallest p among them: 0.115 (Saturn retrograde, 1.02×)	≥0.11	None pass
1964–1989	Sun in Viśākhā; Moon in the sharp star group	233 vs 221.7 (1.05×); 903 vs 905.2 (1.00×)	0.23; 0.54	None pass
1900–1963, M7.0+ (670)	Moon within 5° of Sun or a planet; the 21 rules	102 vs 99.0 (1.03×); smallest rule p 0.079	0.39 (time-shift 0.36); ≥0.079	None pass

The data-first scan repeated

Data-first scan. We recomputed all 433 sky bins on the 6,111 earthquakes of 1964–1989 and compared them with the 1990–2026 scan. If the 1990–2026 patterns were real, the two lists of bin scores would agree. The agreement is r=−0.008 (chance level is −0.003±0.064; p=0.54). Of the eight strongest 1990–2026 bins, none was strong in the same direction in 1964–1989, and no bin reached |z|≥2.5 in both eras.

Test (frozen in advance)	Data scored	Result	Verdict
Data-first scan: do the 433 bin scores of 1990–2026 reappear?	1964–1989	r=−0.008, p=0.54; no bin strong in both eras	Not replicated

TG-SAI. The corrected TG-SAI model was also scored once, without retraining, on Older set 1, Older set 2 and the Repository file (AUC 0.50 to 0.51, no skill). The details are in the Rule-First Study, section 9.3.

What this adds. The data now span 1900–2026, more than a century, and cover 1964–1989 at small sizes. A pattern that was real would have shown up somewhere in this span. None did.

7. Stage 5: bins defined by the source texts (all treatises)

The first four stages scanned generic sky bins. Stage 5 adds every bin that needs a text to define it, so that the Brihat Saṃhitā (Sanskrit and English), the Adbhuta Sāgara and the Parāśara Tantra are all searched from the data side too. The 54 bins were fixed in the script before the first run (preregistration_data_first_treatise_scan.md). Nothing is told which way an effect should point: each bin gets a two-sided z-score against a matched background of 6-hourly sky positions, with a random 0–6 h shift and the earthquake locations shuffled onto the background times. Bins are found in 1990–2007 and checked in 2008–2026 (Modern M5.5+ set); the same bins are then read in Older set 1 (1964–1989) and Older set 2 (1900–1963). Bonferroni bar: 0.05 ÷ 54 = 0.00093.

Bin family	Bins	Source
Moon's nakṣatra circle under eight lists (Vāyu, Agni, Indra, Varuṇa)	32	Parāśara Tantra (corrected; and as quoted in the Adbhuta Sāgara), Brihat Saṃhitā (Sanskrit; English readings 1–3), Adbhuta Sāgara (Bṛhaspati's list; compiler's summary)
Yāma-clock circle (night reversed; night in the same order)	8	Adbhuta Sāgara
Yāma circle equals the Moon's circle (two lists × two night orders)	4	Adbhuta Sāgara, Brihat Saṃhitā
Clock-time circle in four 6-hour blocks	4	Brihat Saṃhitā chapter 32
Moon in one of the four disputed nakṣatras; twilight window; eclipse in the 7 days before	3	Adbhuta Sāgara; Brihat Saṃhitā chapter 32
Mercury–Jupiter within 1° (also with Mercury the southern one); any two star-planets within 1°	3	Brihat Saṃhitā chapter 17 (planetary war)

Result. 6 of the 54 bins reached p<0.05 in 1990–2007 (about 2.7 expected by chance). All six are one effect counted six times: the Moon in the Indra circle was about 7% below expectation (955 vs 1,033 with the Brihat Saṃhitā Sanskrit list), whichever treatise's list is used, because the lists share most nakṣatras. None passes the Bonferroni bar, and in 2008–2026 the same bins sit at z between −0.2 and 0.0 (1,045 vs 1,047). Replication count: 0 of 6, against 0.15 expected by chance.

Bin	1990–2007 z	2008–2026 z	Older set 1 z	Older set 2 z
Moon in the BS Sanskrit circle: Indra	-2.76	-0.08	+0.86	-1.73
Moon in the BS English v1 circle: Indra	-2.70	+0.03	+0.72	-1.70
Moon in the AS compiler circle: Indra	-2.61	-0.13	+0.91	-1.62
Moon in the BS English v3 circle: Indra	-2.53	-0.17	+0.52	-1.36
Moon in the BS English v2 circle: Indra	-2.43	-0.22	+0.45	-1.25
Moon in the AS Brihaspati circle: Indra	-2.18	+0.02	+0.65	-1.27
Mercury-Jupiter within 1 deg (BS 17 war)	+0.44	+3.77	-0.76	+1.58

The one bin that passes in the holdout is not a discovery. Mercury within 1° of Jupiter (Brihat Saṃhitā 17.16) has 19 events against 8.2 expected in 2008–2026 (z=+3.77, past the 0.00093 bar), but in 1990–2007 it shows 9 against 7.8 (z=+0.44), so it was not found first and then confirmed. It is the same result as in the Rule-First Study: the events are clustered in a few episodes, the conjunction-level test gives p=0.022 against a bar of 0.0125, and Older sets 1 and 2 do not repeat it (z=−0.76 and +1.58).

Do the eras agree with each other? If a real pattern existed, the 54 z-scores of 1990–2007 would resemble those of the other periods. Because the bins overlap, chance was measured by redrawing events from the background 3,000 times and recomputing the whole scan:

Compared with 1990–2007	Correlation of z-scores	Chance spread (sd)	p (two-sided)
2008–2026 (the pre-set check)	+0.14	0.30	0.668
Older set 1 (1964–1989)	-0.36	0.31	0.281
Older set 2 (1900–1963)	+0.81	0.31	0.001

What this means. The pre-set check (2008–2026) shows no agreement, and Older set 1 points the opposite way. Older set 2 (M7+, 1900–1963) does resemble 1990–2007 (r=+0.81, p=0.001, or 0.003 after allowing for the three comparisons). We could not trace this to a single bin: it persists when any one family of bins is removed. A real effect would also appear in the 2008–2026 half of the same catalogue, which it does not, so it is recorded as an unexplained coincidence to test on a fourth, independent period rather than as support for any treatise. Chapters 3, 5, 17 and 24 of the Brihat Saṃhitā exist here only in English; all other chapters used for these bins were read in Sanskrit.

8. Related tests in the Rule-First Study

Five further families of tests were run on these same data sets and are reported in the Rule-First Study, which holds their method and tables. All are null; they are listed here so that this report is complete.

Test	Data	Outcome	Where
Brihat Saṃhitā read from the Sanskrit and from the English translation: four circles, time of day, felt distance, post-eclipse window (50 tests, bar 0.005)	Modern M5.5+ set, Older sets 1 and 2	Four-circle counts within 2–3% of chance. One 16-pair test in Older set 1 reached p=0.0048 but failed in the other sets.	Rule-First Study, section 6.4

Test	Data	Outcome	Where
Crescent-Moon 'carriage pole' omen computed for a true horizon	All four periods	1.03×, 0.97×, 0.95× (Ujjain reading); epicentre reading 1.04× (p=0.019) on the 2008+ holdout, 1.00× and 1.02× elsewhere. Not supported.	Rule-First Study, section 7
Adbhuta Sagara (third treatise): four circles with three authority lists, the yāma clock, twilight and felt distance (13 tests, bar 0.0038)	Modern M5.5+ set (all, 1990–2007, 2008+), Older sets 1 and 2	None passes (smallest p=0.072). A grid artifact that briefly looked like a twilight deficit and an Older set 1 hit was removed by shifting background times.	Rule-First Study, section 6.5
Rest of the Brihat Saṃhitā: whole English text searched for earthquake rules; Chapter 17.16 (Mercury defeated by Jupiter) tested at event and conjunction level	Modern M5.5+ set (all, 1990–2007, 2008+), Older sets 1 and 2	Not supported. Event counts look high (28 vs 16.2) but come from clustered shocks in 14 of 20 conjunctions; per conjunction p=0.052 overall and 0.022 in the holdout, neither past the bar of 0.0125; no replication.	Rule-First Study, section 6.6
Regional belts (6 belts × 7 rules = 42 tests per period, bar 0.0012)	All four periods	None passes. A faint crescent pattern in Japan and Indonesia (1.19× pooled; 0.06 after correction) is recorded as descriptive only.	Rule-First Study, section 8

9. Conclusion

Verdict. Starting from the data instead of the rules, and using the Modern M6.5+ set, the Modern M5.5+ set (9,368 earthquakes, 1990–2026), Older set 1 and Older set 2, we found **no star group, Moon phase, planetary arrangement, mandala, gap factor or treatise-defined bin from the Parāśara Tantra, the Brihat Saṃhitā or the Adbhuta Sāgara that is reliably linked to earthquakes**. The two leads from the Modern M6.5+ set (Moon in the sharp group; Sun in Viśākhā) vanished on 8,098 fresh events (0.98× and 1.02× the expected counts). None of the 23 rules passes the strict bar, and none of the gap tests does either.

Why this is a solid 'no'. With this many events, an excess as small as the 17% first seen in the sharp group would have been caught, so it is ruled out. The rules and thresholds were written down before testing, and results were confirmed on data not used to choose them.

The last lead is closed. The one faint lead, the Moon within 5° of the Sun or a planet in small earthquakes (p=0.0030 in the Modern M5.5+ set), was tested on Older set 1 and Older set 2 with the rule written down first and showed no excess (0.98×, p=0.73). The data-first scan repeated on Older set 1 and the TG-SAI model scored on older data also showed nothing. **What is still open.** A faint, uncorrected crescent pattern in Japan and Indonesia needs new earthquakes to test fairly (Rule-First Study, section 8). Things the records cannot test (omens, haloes, named stars, effects on people) remain untested, not disproved.

Plain-language takeaway. Make the sample ten times bigger and the apparent patterns disappear. That is what chance clusters do, and it is the main reason to be careful when a single result looks exciting.

Limits

• Only the variables that were binned could be found; effects tied to place and season, or to combinations not tested, were not searched (section 5 covers many combinations). • Stage 5 uses the lists as the texts and our readings give them: the Brihat Saṃhitā Sanskrit covers printed pages 393–683 only, so chapters 3, 5, 17 and 24 are English only; the 54 treatise bins are strongly overlapping (six lists share most nakṣatras), so the Bonferroni bar of 0.00093 is conservative. • Bins overlap (the Sun's nakṣatra, sign and separations share dates), so the 433 bins are not independent and r is only a summary. • Lahiri ayanāṃśa only; other systems move the Moon and Sun between bins (Rule-First Study, section 11). • Sky positions for the Modern M5.5+ set and older sets come from our own formulas, validated against the ephemeris tool, not the tool itself. • Declustering removes most but not all aftershocks; the time-shift test guards against the remainder. • Catalogue completeness: M5.5+ is complete since 1990; Older set 1 is less complete for small events; Older set 2 has no depth data. • Small p-values found by splitting the data after the fact (such as by size) are descriptive, not proof. • The half-split of the Modern M6.5+ set costs power: each half has about 630 events.

10. Files

Stage 1: bottom_up_bin_scan.csv, preregistration_bottom_up_followup.md, sun_nakshatra_1964_1989_followup.csv, prereg_slow_planet_recheck_matched_baseline.csv, declustered_events_M65.csv. **Stage 5 (all treatises):** preregistration_data_first_treatise_scan.md, data_first_treatise_bin_scan.csv, data_first_treatise_scan_agreement.csv. **Stage 2:** preregistration_m55_replication.md, preregistration_adbhuta_sagara.md, adbhuta_sagara_results.csv, preregistration_brihat_samhita_all_chapters.md, brihat_samhita_ch17_mercury_jupiter_results.csv,

quakes_m55_input.csv (9,368 events), m55_primary_tests.csv, m55_23rules_results.csv, m55_by_magnitude_band.csv, m55_bottom_up_bin_scan_new_events.csv. **Gap tests:** preregistration_gap_tests.md, gap_tests_primary.csv, gap_tests_two_condition_screen.csv, gap_slow_planet_time_shift_check.csv. **Older eras:** preregistration_historical_test.md, preregistration_tgsai_bottomup_historical.md, historical_replication_results.csv, historical_bin_scan_comparison.csv, historical_eraA_1964_1989_events.csv, historical_eraB_1900_1963_events.csv.

Terms in plain language

Term	Plain meaning
Nakṣatra	One of 27 star groups along the Moon's path across the sky, about 13° wide each.
Mandala (circle)	A group of nakṣatras that a text assigns to one of four winds or deities (Vāyu, Agni, Indra, Varuṇa); each text gives its own list.
Yāma	One of four equal parts of the day (or night) in the Adbhuta Sāgara, each tied to one circle; a clock version of the mandala idea.
Sharp group (Tīkṣṇa)	Four nakṣatras (Ārdrā, Āśleṣā, Jyeṣṭhā, Mūla) that the Parāśara Tantra calls harsh.
Viśākhā	The 16th nakṣatra; the Sun is in it from about 16 to 29 November.
Declustering	Removing aftershocks so each earthquake counts once.
Pre-registration	Writing the test and pass mark down before looking at the data.
Discovery and holdout	Two halves of the data: before 2008 (to look for effects) and from 2008 (to confirm them).
Bonferroni correction	Making the pass mark stricter when many tests are run (0.05 ÷ number of tests).
Time-shift test	Moving every earthquake by the same random number of days: clusters stay, the sky link breaks. It shows what chance looks like.
Replicate	Repeat on fresh data and get the same result.
AUC	Area under the ROC curve: 0.5 means a model cannot tell real cases from random ones; 1.0 is perfect.
z-score	How many standard deviations a count lies above or below what chance predicts.

Appendix: the 23 rules in full

These are the 23 pre-registered rules of the Rule-First Study, each as one test. Each row names its source chapter (verse numbers are given only where they were read and checked in the Sanskrit). This first family of rules comes from the Parāśara Tantra (earthquake chapter 18, Sun chapter 3, Rāhu and eclipses chapter 5, planets chapters 6–10, nakṣatra and tithi chapters 14–16) and from simple sky features; the Brihat Saṃhitā and Adbhuta Sāgara rules were frozen later as separate families (Rule-First Study, section 3). Each was tested on the 4,707 earthquakes of the Modern M5.5+ set from 2008 onward, after being tried on the 4,661 earlier ones. A rule passes only if p is below 0.0022 (0.05 ÷ 23) and the pattern also shows before 2008. **None passes.** Where several numbers are shown (for example 1059/1407/1232/1009), they are counts per group seen versus expected.

Rule	Source and chapter	What it tests (technical)	Plain meaning	Seen vs expected, 2008+	p-value
R1 (stars)	Parāśara Tantra ch.18, verses 18.3–18.6 (checked against the Sanskrit)	Moon's star group sets one of four earthquake types: Anila, Anala, Indra, Varuṇa (Parāśara Tantra ch.18)	Are the four types seen in unequal numbers?	1059/1407/1232/1009 vs 1048/1394/1221/1044	0.684
R1 (season)	Same verses; tropical marking is my variation	Same four types, with star groups marked against the seasons (tropical)	Same question with the other way of marking the sky.	1065/1369/1220/1053 vs 1041/1395/1224/1047	0.782
R2 (stars)	Parāśara Tantra ch.14 (nakṣatra groups); chapter cited, verse not recorded	Moon in Ārdrā, Āśleṣā, Jyeṣṭhā or Mūla, the 'sharp' group (ch.14)	More quakes when the Moon is in the harsh star groups?	692 vs 697.6	0.597
R2 (season)	Same chapter; tropical variation	Same sharp-group rule, tropical marking	Same, other marking.	719 vs 698.9	0.211
R3	Parāśara Tantra ch.16 (tithi types, Riktā)	Moon in lunar day 4, 9 or 14 of each 15-day cycle, Riktā or 'empty' (ch.16)	More quakes on the 'empty' lunar days?	949 vs 945.1	0.449
R4	TG-SAI repository (syzygy feature); tithi background in Parāśara Tantra ch.16	Within one lunar day of new or full moon	More quakes near new or full moon, when tides are strongest?	955 vs 924.7	0.138

Rule	Source and chapter	What it tests (technical)	Plain meaning	Seen vs expected, 2008+	p-value
R5	Parāśara Tantra ch.5 (Rāhu, eclipses), esp. 5.4 and 5.7	A solar or lunar eclipse within 15 days (Rāhu chapter 5)	More quakes near an eclipse?	1305 vs 1265.5	0.100
R6	Parāśara Tantra ch.5, as R5	An eclipse within 7 days	Same, tighter window.	835 vs 804.7	0.124
R7 Mars	Parāśara Tantra ch.6 (Mars)	Mars appears to move backwards (chapter 6)	More quakes while Mars is retrograde?	436 vs 426.8	0.327
R7 Mercury	Parāśara Tantra ch.7 (Mercury)	Mercury retrograde (ch.7)	Same, Mercury.	890 vs 903.2	0.693
R7 Jupiter	Parāśara Tantra ch.8 (Jupiter)	Jupiter retrograde (ch.8)	Same, Jupiter.	1372 vs 1396.9	0.791
R7 Venus	Parāśara Tantra ch.9 (Venus)	Venus retrograde (ch.9)	Same, Venus.	335 vs 336.6	0.545
R7 Saturn	Parāśara Tantra ch.10 (Saturn)	Saturn retrograde (ch.10)	Same, Saturn.	1758 vs 1774.7	0.697
R8 (stars)	Parāśara Tantra ch.6–10 (Saturn–Jupiter in Kṛttikā/Viśākhā); simplified to 'same star group'	Jupiter and Saturn in the same star group (ch.10)	More quakes when the two slow giants share a star group?	162 vs 171.6	0.782
R8 (season)	Same; tropical variation	Same rule, tropical marking	Same, other marking.	132 vs 137.9	0.706
R9	Parāśara Tantra ch.15 (planetary wars and formations)	Any two of Mars, Mercury, Jupiter, Venus, Saturn within 3° of each other, a 'planetary war' (ch.15)	More quakes when two planets are very close?	1019 vs 986.8	0.129
R10	Parāśara Tantra ch.14 (Moon–planet approach afflicting nakṣatras)	Moon within 5° of the Sun or a planet (ch.14)	More quakes when the Moon sits next to a planet or the Sun?	756 vs 688.2	0.003
R11 (stars)	Parāśara Tantra ch.3 (Sun; six ṛtus from the winter solstice)	Sun's position in the six ṛtus, 60° seasons from the winter solstice (ch.3)	Do quakes favour certain seasons?	766/819/800/807/788/727 vs 781/805/817/803/750/751	0.613
R11 (season)	Same; tropical variation	Same six seasons, tropical marking	Same, other marking.	735/798/811/827/767/769 vs 761/785/809/816/797/738	0.590
R12 magnitude (stars)	Mandalas from Parāśara Tantra 18.3–18.6; magnitude comparison is our addition	Do the four types differ in earthquake magnitude?	Are some types stronger?	H=7.44 (groups compared, no difference)	0.059
R12 depth (stars)	As above; depth comparison is our addition	Do the four types differ in depth?	Are some types deeper?	H=0.80 (groups compared, no difference)	0.850
R12 magnitude (season)	As above; tropical variation	Same magnitude comparison, tropical marking	Same, other marking.	H=1.80 (groups compared, no difference)	0.614
R12 depth (season)	As above; tropical variation	Same depth comparison, tropical marking	Same, other marking.	H=0.91 (groups compared, no difference)	0.822

How to read the p-value column. p is the chance of a result this strong if the sky and earthquakes were unrelated. Values near 0.5 or higher mean the data look like chance. The lowest here is 0.003 (rule R10, section 3.4), which misses the pass mark of 0.0022 and did not replicate on older data (section 6).